Lesson 37: Learning in Games and Bayesian Games

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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5 Zero-Sum Games: Fictitious Play and No-Regret Learning

We return to two-player zero-sum games $G = (\{1,2\}, A_1, A_2, g)$ where $g_1(a) = -g_2(a) = g(a)$. Recall from Lesson 11 the minimax theorem: $\max_{\alpha_1} \min_{\alpha_2} \mathbb{E}[g(\alpha_1, \alpha_2)] = \min_{\alpha_2} \max_{\alpha_1} \mathbb{E}[g(\alpha_1, \alpha_2)] = v^*$ (the value of the game).

5.1 Fictitious Play

Definition 5.1 (Fictitious Play)**.** *Fictitious play* is a learning dynamic where each player best-responds to the empirical frequency of the opponent's past actions. Formally, let $a_j^1, a_j^2, \ldots$ be the actions played by player j. The empirical frequency after T rounds is

$$\hat{\alpha}_j^T(a_j) = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{a_j^t = a_j\} \qquad \text{for each } a_j \in A_j.$$

At round $T+1$, player i plays $a_i^{T+1} \in \mathrm{BR}_i(\hat{\alpha}_j^T)$, a best response to the empirical distribution of the opponent.

Theorem 5.1 (Convergence of Fictitious Play in 2×2 Zero-Sum Games)**.** *In any 2×2 zero-sum game, the empirical frequencies $(\hat{\alpha}_1^T, \hat{\alpha}_2^T)$ under fictitious play converge to a Nash equilibrium as $T \to \infty$.*

Proof. Let the payoff matrix be

$$G = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with player 1 choosing rows and player 2 choosing columns, so $g_1 = G$, $g_2 = -G$.

Consider the function $\Phi^T = \max_{a_1} \sum_{t=1}^{T} g(a_1, a_2^t)$. This is player 1's "fictitious play payoff"—the best she could have done against the sequence of player 2's actions if constrained to a fixed action.

We analyze how Φ^T evolves. Note that $\Phi^T = \max\{R_1^T, R_2^T\}$ where $R_k^T = \sum_{t=1}^{T} g(k, a_2^t)$ is the cumulative payoff of row k.

In a 2×2 game, the best response of player 1 to $\hat{\alpha}_2^T$ is row 1 if $R_1^T > R_2^T$ and row 2 if $R_1^T < R_2^T$ (with either at ties). Define $D^T = R_1^T - R_2^T$.

Claim: $|D^T|$ is bounded. To see this, note that when $D^T > 0$, player 1 plays row 1, and player 2 best-responds to the empirical frequency $\hat{\alpha}_1^T$ which puts more weight on row 1. Player 2's best response tends to be the column that makes row 1 less attractive, which decreases D^{T+1}. Similarly when $D^T < 0$. The key observation in the 2×2 case is that D^T oscillates around 0 with bounded amplitude (bounded by $\max |g(a)|$), because each step changes D^T by at most $\max_{a_2} |g(1, a_2) - g(2, a_2)|$.

More precisely, define $V_1^T = \frac{1}{T}\Phi^T$ (player 1's "best fixed action" average payoff). We have

$$V_1^T = \frac{1}{T} \max_{a_1} \sum_{t=1}^{T} g(a_1, a_2^t) \geq \frac{1}{T} \sum_{t=1}^{T} g(a_1^t, a_2^t),$$

since the best fixed action does at least as well as the actual sequence.

Similarly, define $V_2^T = \frac{1}{T} \max_{a_2} \sum_{t=1}^{T} (-g(a_1^t, a_2))$ for player 2. By the minimax theorem for the stage game:

$$V_1^T \geq v^* \geq -V_2^T.$$

Furthermore, $V_1^T - v^* \to 0$ and $V_2^T + v^* \to 0$ because the "regret" $V_1^T - \frac{1}{T}\sum_t g(a_1^t, a_2^t)$ can be bounded by the bounded oscillation of $D^T/T \to 0$.

Since $|D^T|$ is bounded, $\frac{D^T}{T} \to 0$, which means $\frac{R_1^T}{T}$ and $\frac{R_2^T}{T}$ converge to the same limit. This implies that $\hat{\alpha}_2^T$ converges to making player 1 indifferent between rows, which is exactly the

NE condition in a 2×2 game with a completely mixed equilibrium. If the game has a pure equilibrium, fictitious play converges to it in finitely many steps.

Thus $(\hat{\alpha}_1^T, \hat{\alpha}_2^T) \to (\alpha_1^*, \alpha_2^*)$, a Nash equilibrium. □

Theorem 5.2 (Robinson's Theorem: Fictitious Play in General Zero-Sum Games)**.** *In any finite two-player zero-sum game, the empirical frequencies under fictitious play converge to the set of Nash equilibria.*

Proof sketch. The full proof (Robinson, 1951) uses a potential function argument. Define

$$\Phi_1^T = \max_{a_1 \in A_1} \sum_{t=1}^{T} g(a_1, a_2^t), \qquad \Phi_2^T = \max_{a_2 \in A_2} \sum_{t=1}^{T} (-g(a_1^t, a_2)).$$

Then $\frac{1}{T}\Phi_1^T \geq v^*$ and $\frac{1}{T}\Phi_2^T \geq -v^*$ (each player's best fixed action does at least as well as the game value against the empirical opponent distribution).

The key step is showing that $\frac{1}{T}(\Phi_1^T + \Phi_2^T) \to 0$. This follows because $\Phi_1^T + \Phi_2^T$ grows sublinearly: when player 1 plays a best response to $\hat{\alpha}_2^T$, the increase in $\Phi_1^{T+1} - \Phi_1^T$ is exactly $g(a_1^{T+1}, a_2^{T+1})$, while the increase $\Phi_2^{T+1} - \Phi_2^T \leq -g(a_1^{T+1}, a_2^{T+1}) + C$ for a constant C related to the number of actions. Carefully tracking these increments shows $\Phi_1^T + \Phi_2^T = O(\sqrt{T})$.

Since $\frac{1}{T}\Phi_1^T \to v^*$ and $\frac{1}{T}\Phi_2^T \to -v^*$, any limit point (α_1^*, α_2^*) of $(\hat{\alpha}_1^T, \hat{\alpha}_2^T)$ satisfies the NE conditions: α_1^* is a best response to α_2^* and vice versa. □

Fictitious Play and Policy-Space Response Oracles (PSRO)

Fictitious play is a precursor to the *PSRO* framework in modern MARL. In PSRO, each player maintains a population of policies (analogous to the empirical frequency in FP), and computes a best response to the current mixture—exactly as in fictitious play, but with neural-network policies instead of tabular strategies. The convergence guarantees of FP in zero-sum games provide theoretical backing for PSRO's effectiveness in competitive MARL.

5.2 No-Regret Learning

Definition 5.2 (External Regret)**.** Player i's *external regret* after T rounds is

$$R_i^T = \max_{a_i \in A_i} \sum_{t=1}^{T} g_i(a_i, a_{-i}^t) - \sum_{t=1}^{T} g_i(a_i^t, a_{-i}^t).$$

A learning algorithm has *no external regret* (or is *Hannan consistent*) if $R_i^T/T \to 0$ as $T \to \infty$.

Definition 5.3 (Multiplicative Weights Update (MWU))**.** Let $\eta > 0$ be a learning rate. Player i maintains weights $w_i^t(a_i)$ for each action $a_i \in A_i$:

$$w_i^1(a_i) = 1 \quad \text{for all } a_i,$$
$$w_i^{t+1}(a_i) = w_i^t(a_i) \cdot \exp\big(\eta \cdot g_i(a_i, a_{-i}^t)\big).$$

The mixed strategy at time t is $\alpha_i^t(a_i) = w_i^t(a_i) / \sum_{a_i'} w_i^t(a_i')$.

Theorem 5.3 (MWU Regret Bound)**.** *If $|g_i(a)| \leq 1$ for all a, then with $\eta = \sqrt{\frac{\ln |A_i|}{T}}$, the multiplicative weights update achieves*

$$R_i^T \leq 2\sqrt{T \ln |A_i|}.$$

In particular, $R_i^T/T \to 0$, so MWU is a no-regret algorithm.

Proof. Define the potential $W^t = \sum_{a_i \in A_i} w_i^t(a_i)$, so $W^1 = |A_i|$.

Upper bound on $\ln(W^{T+1}/W^1)$:

$$\begin{aligned}\frac{W^{t+1}}{W^t} &= \frac{\sum_{a_i} w_i^t(a_i)\exp(\eta \cdot g_i(a_i, a_{-i}^t))}{W^t} \\ &= \sum_{a_i} \alpha_i^t(a_i)\exp(\eta \cdot g_i(a_i, a_{-i}^t)) \\ &\le \sum_{a_i} \alpha_i^t(a_i)\big(1 + \eta g_i(a_i, a_{-i}^t) + \eta^2 g_i(a_i, a_{-i}^t)^2\big),\end{aligned}$$

using $e^x \le 1 + x + x^2$ for $|x| \le 1$ (valid when $\eta \le 1$ and $|g_i| \le 1$). Therefore:

$$\begin{aligned}\frac{W^{t+1}}{W^t} &\le 1 + \eta \sum_{a_i} \alpha_i^t(a_i) g_i(a_i, a_{-i}^t) + \eta^2 \\ &\le \exp\big(\eta \cdot g_i(\alpha_i^t, a_{-i}^t) + \eta^2\big),\end{aligned}$$

where $g_i(\alpha_i^t, a_{-i}^t) = \mathbb{E}_{a_i \sim \alpha_i^t}[g_i(a_i, a_{-i}^t)]$ and we used $1 + x \le e^x$. Telescoping:

$$\ln \frac{W^{T+1}}{W^1} \le \eta \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) + \eta^2 T.$$

Lower bound on $\ln(W^{T+1}/W^1)$: For any fixed action a_i^*:

$$W^{T+1} \ge w_i^{T+1}(a_i^*) = \exp\Big(\eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t)\Big).$$

Therefore:

$$\ln \frac{W^{T+1}}{W^1} \ge \eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \ln |A_i|.$$

Combining:

$$\eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \ln |A_i| \le \eta \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) + \eta^2 T.$$

Rearranging:

$$\sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) \le \frac{\ln |A_i|}{\eta} + \eta T.$$

Since this holds for all a_i^*, taking the max and substituting $\eta = \sqrt{\frac{\ln |A_i|}{T}}$:

$$R_i^T \le \sqrt{T \ln |A_i|} + \sqrt{T \ln |A_i|} = 2\sqrt{T \ln |A_i|}.$$ □

Theorem 5.4 (No-Regret Implies NE in Zero-Sum Games)**.** *In a two-player zero-sum game, if both players use no-regret learning algorithms, then the time-averaged strategy profile $(\bar{\alpha}_1^T, \bar{\alpha}_2^T)$ converges to the set of Nash equilibria as $T \to \infty$, where $\bar{\alpha}_i^T = \frac{1}{T}\sum_{t=1}^{T} \alpha_i^t$.*

Proof. Let $g = g_1 = -g_2$. The no-regret condition for player 1 gives:

$$\max_{a_1} \frac{1}{T} \sum_{t=1}^{T} g(a_1, \alpha_2^t) \le \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) + \frac{R_1^T}{T}.$$

By linearity of g in the first argument and Jensen's inequality (or directly by linearity):

$$\max_{a_1} g(a_1, \bar{\alpha}_2^T) = \max_{a_1} \frac{1}{T} \sum_{t=1}^{T} g(a_1, \alpha_2^t) \leq \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) + \varepsilon_1^T,$$

where $\varepsilon_1^T = R_1^T/T \to 0$.

Similarly, the no-regret condition for player 2 (who minimizes g) gives:

$$\min_{a_2} g(\bar{\alpha}_1^T, a_2) = \min_{a_2} \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, a_2) \geq \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) - \varepsilon_2^T,$$

where $\varepsilon_2^T \to 0$.

Combining:

$$\min_{a_2} g(\bar{\alpha}_1^T, a_2) \geq \frac{1}{T} \sum_{t} g(\alpha_1^t, \alpha_2^t) - \varepsilon_2^T \geq \max_{a_1} g(a_1, \bar{\alpha}_2^T) - \varepsilon_1^T - \varepsilon_2^T.$$

But by the minimax inequality (which always holds):

$$\max_{a_1} g(a_1, \bar{\alpha}_2^T) \geq \min_{a_2} g(\bar{\alpha}_1^T, a_2).$$

Therefore:

$$0 \leq \max_{a_1} g(a_1, \bar{\alpha}_2^T) - \min_{a_2} g(\bar{\alpha}_1^T, a_2) \leq \varepsilon_1^T + \varepsilon_2^T \to 0.$$

This means the "duality gap" converges to zero. Since $\max_{a_1} g(a_1, \bar{\alpha}_2^T) \to v^*$ and $\min_{a_2} g(\bar{\alpha}_1^T, a_2) \to v^*$, any limit point (α_1^*, α_2^*) of $(\bar{\alpha}_1^T, \bar{\alpha}_2^T)$ satisfies $\max_{a_1} g(a_1, \alpha_2^*) = \min_{a_2} g(\alpha_1^*, a_2) = v^*$, which is the Nash equilibrium condition. □

No-Regret Learning in Multi-Agent RL

No-regret algorithms are fundamental to competitive MARL. The convergence guarantee of Theorem 5.4 underpins algorithms like:

- **Counterfactual Regret Minimization (CFR)** for extensive-form games (poker AI).
- **Online mirror descent** in policy space for continuous games.
- **V-learning** and other model-free algorithms that achieve sublinear regret against adversarial opponents.

The $O(\sqrt{T})$ regret rate of MWU translates to $O(1/\sqrt{T})$ convergence to NE in zero-sum games, which can be accelerated by optimistic variants.

6 Bayesian Games (Incomplete Information)

6.1 Motivation and Model

In many strategic settings, players are uncertain about others' payoff functions. A firm does not know a competitor's cost structure; an agent does not know an opponent's reward function. Harsanyi's (1967–68) framework models this via *types*.

Definition 6.1 (Bayesian Game)**.** A *Bayesian game* (or game of incomplete information) is a tuple $\Gamma = (N, (A_i)_{i \in N}, (\Theta_i)_{i \in N}, p, (u_i)_{i \in N})$ where:

(a) $N = [n]$ is the player set.

(b) A_i is the finite action set of player i.

(c) Θ_i is the finite *type space* of player i. A type $\theta_i \in \Theta_i$ encodes player i's private information.

(d) $p \in \Delta(\Theta)$ is the *common prior* distribution over the type profile $\theta = (\theta_1, \ldots, \theta_n) \in \Theta = \prod_i \Theta_i$.

(e) $u_i : A \times \Theta \to \mathbb{R}$ is player i's payoff function, depending on both the action profile and the type profile.

The timing is: Nature draws $\theta \sim p$; each player i observes only her own type θ_i; players simultaneously choose actions.

Definition 6.2 (Bayesian Nash Equilibrium)**.** A *(pure) strategy* in a Bayesian game is a function $s_i : \Theta_i \to A_i$ mapping types to actions. A strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a *Bayesian Nash equilibrium* (BNE) if, for every player i and every type $\theta_i \in \Theta_i$:

$$s_i^*(\theta_i) \in \operatorname{argmax}_{a_i \in A_i} \mathbb{E}_{\theta_{-i} \sim p(\cdot|\theta_i)} \big[u_i\big(a_i, s_{-i}^*(\theta_{-i}), \theta_i, \theta_{-i}\big) \big],$$

where $p(\theta_{-i}|\theta_i) = p(\theta_i, \theta_{-i})/p(\theta_i)$ is the conditional distribution of others' types given θ_i.

Remark 6.1. The BNE condition says: each type of each player best-responds to the expected play of opponents, where the expectation is over opponents' types (which determine their actions through the strategy profile).

6.2 Existence of BNE

Theorem 6.1 (Existence of BNE)**.** *Every finite Bayesian game (finite player set, finite action sets, finite type spaces) has a Bayesian Nash equilibrium in mixed strategies.*

Proof. Transform the Bayesian game into the "agent normal form." For each player-type pair (i, θ_i), create an "agent" who chooses action $a_i \in A_i$. This yields a finite normal-form game with $\sum_i |\Theta_i|$ agents, each with action set A_i. The payoff to agent (i, θ_i) when the action profile is $(a_{j,\theta_j})_{j,\theta_j}$ is:

$$\sum_{\theta_{-i}} p(\theta_{-i}|\theta_i) \cdot u_i(a_{i,\theta_i}, (a_{j,\theta_j})_{j \neq i}, \theta_i, \theta_{-i}).$$

This is a finite normal-form game, so by Nash's theorem (from Lesson 11), it has a Nash equilibrium in mixed strategies. Any NE of the agent normal form corresponds to a BNE of the original Bayesian game: the mixed strategy of agent (i, θ_i) specifies $s_i(\theta_i) \in \Delta(A_i)$, and the NE condition for each agent is exactly the BNE condition for type θ_i of player i. □

6.3 Examples

Example 6.1 (Cournot Duopoly with Unknown Costs)**.** Two firms produce a homogeneous good with inverse demand $P(Q) = a - Q$ where $Q = q_1 + q_2$. Firm i's cost is $c_i q_i$. Suppose c_1 is known to both firms, but c_2 is private information of firm 2: $c_2 \in \{c_L, c_H\}$ with $\mathbb{P}(c_2 = c_L) = \pi$.

This is a Bayesian game with $\Theta_2 = \{c_L, c_H\}$, $\Theta_1 = \{c_1\}$ (trivial type space for firm 1). Actions are quantities $q_i \geq 0$.

Firm 2's profit when type is $\theta_2 = c$: $u_2 = (a - q_1 - q_2 - c)q_2$.

In a BNE, firm 2 uses strategy $s_2 : \{c_L, c_H\} \to \mathbb{R}_+$ and firm 1 chooses q_1. The first-order conditions give:

$$\begin{aligned} q_1 &= \frac{a - c_1 - \pi s_2(c_L) - (1-\pi)s_2(c_H)}{2}, \\ s_2(c_L) &= \frac{a - c_L - q_1}{2}, \\ s_2(c_H) &= \frac{a - c_H - q_1}{2}. \end{aligned}$$

Solving this system of three equations yields the unique BNE. The key insight is that firm 1's optimal quantity depends on the *expected* cost of firm 2, and firm 2's strategy conditions on its private type.

Example 6.2 (First-Price Sealed-Bid Auction)**.** Two bidders have private values $v_i \sim \text{Uniform}[0,1]$ independently. Each bidder submits a sealed bid b_i; the highest bidder wins and pays her bid. Payoff: $u_i = (v_i - b_i)\mathbf{1}\{b_i > b_j\}$.

This is a Bayesian game with $\Theta_i = [0,1]$ (continuous types). We seek a symmetric BNE where $b_i = \beta(v_i)$ for some increasing function β.

Given opponent's strategy β, bidder i with value v_i solves:

$$\max_b (v_i - b) \cdot \mathbb{P}(\beta(v_j) < b) = (v_i - b) \cdot \beta^{-1}(b).$$

The first-order condition (substituting $b = \beta(v_i)$) and the boundary condition $\beta(0) = 0$ yield $\beta(v_i) = v_i/2$: each bidder bids half her value.

Verification: If bidder j bids $v_j/2$, then bidder i's problem is $\max_b (v_i - b) \cdot 2b = \max_b 2bv_i - 2b^2$. The FOC gives $2v_i - 4b = 0$, so $b = v_i/2$, confirming the BNE.

Bayesian Games and Opponent Modeling in MARL

In MARL, agents typically do not know each other's reward functions—precisely the setting of Bayesian games. The Bayesian framework formalizes *opponent modeling*:

- The type θ_j can encode opponent j's policy parameters or reward weights.
- The prior $p(\theta_{-i})$ represents agent i's beliefs about opponents.
- Bayesian updating (observing opponents' actions and updating beliefs via Bayes' rule) is the principled approach to learning about opponents.

Algorithms like LOLA (Learning with Opponent-Learning Awareness) and Bayesian policy reuse implement variants of this Bayesian framework.